Behavior of The Direct Extension of ADMM

Based on the paper by: Chen, He, Ye, Yuan - 2014

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Introduction

ADMM (Alternating Direction Method of Multipliers)

Traditional ADMM:

$$\begin{aligned} &\min \theta_1(x_1) + \theta_2(x_2) \\ &\text{s.t. } A_1x_1 + A_2x_2 = b, \\ &x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2. \end{aligned} \tag{1}$$

- θ_i are closed convex functions
- $\mathcal{X}_i$ are closed convex sets

Augmented Lagrangian:

$$\mathcal{L}_A(x_1, x_2, \lambda) = \sum_{i=1}^{2} \theta_i(x_i) - \lambda^T(\sum_{i=1}^{2} A_ix_i - b) + \frac{\beta}{2} \| \sum_{i=1}^{2} A_ix_i - b\|^2 \tag{2}$$

Update scheme:

$$\begin{cases} x_1^{k+1} & = \arg\min \left\{\mathcal{L}_A(x_1, x_2^k, \lambda^k) \mid x_1 \in \mathcal{X}_1\right\}, & \text{(3a)} \\ x_2^{k+1} & = \arg\min \left\{\mathcal{L}_A(x_1^{k+1}, x_2, \lambda^k) \mid x_2 \in \mathcal{X}_2\right\}, & \text{(3b)} \\ \lambda^{k+1} & = \lambda^k - \beta(A_1x_1^{k+1} + A_2x_2^{k+1} - b). & \text{(3c)} \end{cases}$$

ADMM (Alternating Direction Method of Multipliers)

Extended ADMM:

$$\begin{aligned} &\min \theta_1(x_1) + \theta_2(x_2) + \theta_3(x_3) \\ &\text{s.t. } A_1x_1 + A_2x_2 + A_3x_3 = b, \\ &x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, x_3 \in \mathcal{X}_3. \end{aligned} \tag{4}$$

- θ_i are closed convex functions
- $\mathcal{X}_i$ are closed convex sets

Augmented Lagrangian:

$$\mathcal{L}_A(x_1, x_2, x_3, \lambda) = \sum_{i=1}^{3} \theta_i(x_i) - \lambda^T(\sum_{i=1}^{3} A_i x_i - b) + \frac{\beta}{2}|| \sum_{i=1}^{3} A_i x_i - b||^2 \tag{5}$$

Update scheme:

$$\begin{aligned} x_1^{k+1} &= \textbf{arg min} \left\{ \mathcal{L}_A(x_1, x_2^k, x_3^k, \lambda^k) \mid x_1 \in X_1 \right\}, \\ x_2^{k+1} &= \textbf{arg min} \left\{ \mathcal{L}_A(x_1^{k+1}, x_2, x_3^k, \lambda^k) \mid x_2 \in X_2 \right\}, \\ x_3^{k+1} &= \textbf{arg min} \left\{ \mathcal{L}_A(x_1^{k+1}, x_2^{k+1}, x_3, \lambda^k) \mid x_3 \in X_3 \right\}, \\ \lambda^{k+1} &= \lambda^k - \beta(A_1x_1^{k+1} + A_2x_2^{k+1} + A_3x_3^{k+1} - b). \end{aligned}$$

Sufficient conditions for convergence

Case 1: $A_1^T A_2 = 0$ or $A_2^T A_3 = 0$

Assume $A_1^T A_2 = 0$

The First Order Conditions for Optimality $\Rightarrow$

$$
\begin{cases}
\theta_1(x_1) - \theta_1(x_1^{k+1}) + (x_1 - x_1^{k+1})^T \left\{ -A_1^T \left[\lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^k + A_3 x_3^k - b) \right] \right\} & \geq 0, \quad \forall x_1 \in \mathcal{X}_1, \quad (6a) \\
\theta_2(x_2) - \theta_2(x_2^{k+1}) + (x_2 - x_2^{k+1})^T \left\{ -A_2^T \left[\lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^k - b) \right) \right\} & \geq 0, \quad \forall x_2 \in \mathcal{X}_2, \quad (6b) \\
\theta_3(x_3) - \theta_3(x_3^{k+1}) + (x_3 - x_3^{k+1})^T \left\{ -A_3^T \left[\lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b) \right] \right\}) & \geq 0, \quad \forall x_3 \in \mathcal{X}_3. \quad (6c)
\end{cases}
$$

Case 1: $A_1^T A_2 = 0$ or $A_2^T A_3 = 0$

Now, $A_1^T A_2 = 0 \Rightarrow$

$$
\begin{cases}
\theta_1(x_1) - \theta_1(x_1^{k+1}) + (x_1 - x_1^{k+1})^T \left\{-A_1^T \left[\lambda^k - \beta(A_1 x_1^{k+1} + \cancel{A_2 x_2^k +} A_3 x_3^k - b)\right]\right\} & \geq 0, \quad \forall x_1 \in \mathcal{X}_1, \quad (7a) \\
\theta_2(x_2) - \theta_2(x_2^{k+1}) + (x_2 - x_2^{k+1})^T \left\{-A_2^T \left[\lambda^k - \beta(\cancel{A_1 x_1^{k+1} +} A_2 x_2^{k+1} + A_3 x_3^k - b)\right)\right\} & \geq 0, \quad \forall x_2 \in \mathcal{X}_2, \quad (7b) \\
\theta_3(x_3) - \theta_3(x_3^{k+1}) + (x_3 - x_3^{k+1})^T \left\{-A_3^T \left[\lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b)\right]\right\}) & \geq 0, \quad \forall x_3 \in \mathcal{X}_3. \quad (7c)
\end{cases}
$$

This is exactly the FOC of the scheme:

$$
\begin{cases}
(x_1^{k+1}, x_2^{k+1}) = \arg\min \left\{\theta_1(x_1) + \theta_2(x_2) - (\lambda^k)^T(A_1 x_1 + A_2 x_2) + \frac{\beta}{2}\|A_1 x_1 + A_2 x_2 + A_3 x_3^k - b\|^2 \mid x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2\right\}, & (8a) \\
x_3^{k+1} = \arg\min \left\{\theta_3(x_3) - (\lambda^k)^T A_3 x_3 + \frac{\beta}{2}\|A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3 - b\|^2 \mid x_3 \in \mathcal{X}_3\right\}, & (8b) \\
\lambda^{k+1} = \lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b). & (8c)
\end{cases}
$$

This is a specific application of the original ADMM, if we consider:

- (x_1, x_2) as one variable
- $[A_1, A_2]$ as one matrix
- and $\theta_1(x_1) + \theta_2(x_2)$ as one function.

So existing convergence results for the original ADMM hold.

Case 2: $A_1^T A_3 = 0$

Note that the update order is cyclical: $x_1 \to x_2 \to x_3 \to \lambda$.
So equivalently, update via: $x_2 \to x_3 \to \lambda \to x_1$.

This gives us the iterative form:

$$\begin{cases} x_2^{k+1} = \arg\min \left\{ \mathcal{L}_A(x_1^k, x_2, x_3^k, \lambda^k) \mid x_2 \in \mathcal{X}_2 \right\}, & \text{(9a)} \\ x_3^{k+1} = \arg\min \left\{ \mathcal{L}_A(x_1^k, x_2^{k+1}, x_3, \lambda^k) \mid x_3 \in \mathcal{X}_3 \right\}, & \text{(9b)} \\ \lambda^{k+1} = \lambda^k - \beta \left(A_1 x_1^k + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b \right), & \text{(9c)} \\ x_1^{k+1} = \arg\min \left\{ \mathcal{L}_A(x_1, x_2^{k+1}, x_3^{k+1}, \lambda^{k+1}) \mid x_1 \in \mathcal{X}_1 \right\}. & \text{(9d)} \end{cases}$$

As with Case 1, by considering the FOC and using $A_1^T A_3 = 0$ we arrive at:

$$\begin{cases} \theta_2(x_2) - \theta_2(x_2^{k+1}) + (x_2 - x_2^{k+1})^T \left\{ -A_2^T \lambda^{k+1} + \beta A_2^T A_3 (x_3^k - x_3^{k+1}) \right\} \geq 0, \quad \forall x_2 \in \mathcal{X}_2, & \text{(10a)} \\ \theta_3(x_3) - \theta_3(x_3^{k+1}) + (x_3 - x_3^{k+1})^T \left\{ -A_3^T \lambda^{k+1} \right\} \geq 0, \quad \forall x_3 \in \mathcal{X}_3, & \text{(10b)} \\ \left(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b \right) + A_1 (x_1^k - x_1^{k+1}) - \frac{1}{\beta} (\lambda^k - \lambda^{k+1}) = 0, & \text{(10c)} \\ \theta_1(x_1) - \theta_1(x_1^{k+1}) + (x_1 - x_1^{k+1})^T \left\{ -A_1^T \lambda^{k+1} - \beta A_1^T A_1 (x_1^k - x_1^{k+1}) + A_1^T (\lambda^k - \lambda^{k+1}) \right\} \geq 0, \quad \forall x_1 \in \mathcal{X}_1. & \text{(10d)} \end{cases}$$

Case 2: $A_1^T A_3 = 0$

Remark: The extended ADMM model (2) can be characterized by a variational inequality, in the sense that finding a saddle point of the Lagrangian (2) is equivalent to solving the following variational problem:

Find $w^* \in \Omega$ such that

$$\mathrm{VI}(\Omega, F, \theta) : \theta(u) - \theta(u^*) + (w - w^*)^T F(w^*) \geq 0, \quad \forall w \in \Omega, \tag{11}$$

where

$$u = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad w = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda \end{pmatrix}, \quad \theta(u) = \theta_1(x_1) + \theta_2(x_2) + \theta_3(x_3),$$

and

$$F(w) = \begin{pmatrix} -A_1^T \lambda \\ -A_2^T \lambda \\ -A_3^T \lambda \\ A_1 x_1 + A_2 x_2 + A_3 x_3 - b \end{pmatrix}.$$

Lemma 2.3.

Let $w^{k+1} = (x_1^{k+1}, x_2^{k+1}, x_3^{k+1}, \lambda^{k+1})$ be generated by (10) from given $v^k = (x_1^k, x_3^k, \lambda^k)$. If $A_1^T A_3 = 0$, we have $\tilde{w}^k \in \Omega$ and

$$\theta(u) - \theta(\tilde{u}^k) + (w - \tilde{w}^k)^T F(\tilde{w}^k) \geq \frac{1}{2}(\|v - v^{k+1}\|_H^2 - \|v - v^k\|_H^2) + \frac{1}{2}\|v^k - v^{k+1}\|_H^2, \quad \forall w \in \Omega,$$

where $\tilde{w}^k$ and $\tilde{u}^k$ are the auxiliary sequences defined below:

$$\tilde{w}^k := \begin{pmatrix} \tilde{x}_1^k \\ \tilde{x}_2^k \\ \tilde{x}_3^k \\ \tilde{\lambda}^k \end{pmatrix} := \begin{pmatrix} x_1^{k+1} \\ x_2^{k+1} \\ x_3^{k+1} \\ \lambda^{k+1} - \beta A_3(x_3^k - x_3^{k+1}) \end{pmatrix}, \quad \text{and} \quad \tilde{u}^k := \begin{pmatrix} \tilde{x}_1^k \\ \tilde{x}_2^k \\ \tilde{x}_3^k \end{pmatrix},$$

Note: the norm in the inequality is with respect to matrix H which is defined as:

$$H = \begin{pmatrix} \beta A_1^T A_1 & 0 & -A_1^T \\ 0 & \beta A_3^T A_3 & 0 \\ -A_1 & 0 & \frac{1}{\beta} I \end{pmatrix}$$

Theorem 2.4.

Assume $A_1^T A_3 = 0$ for the extended ADMM model. Let $\{x_1^k, x_2^k, x_3^k, \lambda^k\}$ be the sequence generated by the direct extension of ADMM (4). Then, we have:

(i) The sequence $\{v^k := (x_1^k, x_3^k, \lambda^k)\}$ is contractive with respect to the solution of $VI(\Omega, F, \theta)$, i.e.,

$$\|v^{k+1} - v^*\|_H^2 \leq \|v^k - v^*\|_H^2 - \|v^k - v^{k+1}\|_H^2.$$

(ii) If the matrices $[A_1, A_2]$ and A_3 are assumed to be full column rank, then the sequence $\{w^k\}$ converges to a KKT point of the model (4).

Sketch of Proof

(i) From Lemma (2.3), we have that

$$\theta(u) - \theta(\tilde{u}^k) + (w - \tilde{w}^k)^T F(\tilde{w}^k) \geq \frac{1}{2}(\|v - v^{k+1}\|_H^2 - \|v - v^k\|_H^2) + \frac{1}{2}\|v^k - v^{k+1}\|_H^2, \quad \forall w \in \Omega,$$

By setting $w = w^*$ (and thus $v = v^*$, $u = u^*$), and rearranging we have:

$$\begin{aligned}\frac{1}{2}(\|v^k - v^*\|_H^2 - \|v^{k+1} - v^*\|_H^2) - \frac{1}{2}\|v^k - v^{k+1}\|_H^2 &\geq \theta(\tilde{u}^k) - \theta(u^*) + (\tilde{w}^k - w^*)^T F(\tilde{w}^k) \\ &\geq \theta(\tilde{u}^k) - \theta(u^*) + (\tilde{w}^k - w^*)^T F(w^*) \\ &\geq 0\end{aligned}$$

Where the second inequality came from the monotonicity of F, and the third inequality comes from the variational inequality ($VI(\Omega, F, \Theta)$ established in (11)).

Now rearranging, we get the following inequality:

$$\|v^k - v^*\|_H^2 \leq \|v^{k+1} - v^*\|_H^2 + \|v^k - v^{k+1}\|_H^2$$

This shows that v^k is contractive with respect to the solution set of $VI(\Omega, F, \Theta)$.

Key Ideas

(ii)

Using the contractive inequality from (i), $\rightarrow$ the sequences $\{A_1x_1^k - \frac{1}{\beta}\lambda^k\}$ and $\{A_3x_3^k\}$ are bounded.

Note that: $A_1x_1^k + A_2x_2^k = \left(A_1x_1^k - \frac{1}{\beta}\lambda^k\right) - \left(A_1x_1^{k-1} - \frac{1}{\beta}\lambda^{k-1}\right) - A_3x_3^k + b$,
$\Rightarrow\{A_1x_1^k + A_2x_2^k\}$ is bounded.

The assumption that $[A_1, A_2, A_3]$ have full column rank $\Rightarrow$ $\{x_1^k\}, \{x_2^k\}, \{x_3^k\}, \{\lambda^k\}$ are bounded.

Thus, there exists a subsequence $\{x_1^{n_k+1}, x_2^{n_k+1}, x_3^{n_k+1}, \lambda^{n_k+1}\}$ that converges to a limit point: $(x_1^\infty, x_2^\infty, x_3^\infty, \lambda^\infty)$.

By further taking limits, it can be shown that $(x_1^\infty, x_2^\infty, x_3^\infty, \lambda^\infty)$ satisfies the $VI(\Omega, F, \Theta)$, and thus is a KKT point of the model.

Finally, it is shown that (by taking limits of subsequences similar to work in class on Wednesday):
$\lim_{k\to\infty} x_1^k = x_1^\infty, \quad \lim_{k\to\infty} x_2^k = x_2^\infty, \quad \lim_{k\to\infty} x_3^k = x_3^\infty, \quad \lim_{k\to\infty} \lambda^k = \lambda^\infty$,
which establishes (ii) of the Theorem. $\square$

Convergence Rates

Rates of Convergence

- Traditional ADMM: $O(1/k)$
  [He, Yuan '12]

- Modified ADMM: (f and g strongly convex, g quadratic) $O(1/k^2)$
  [Goldstein et al. '12]

- Extended ADMM: ergodic convergence rate of $O(1/k)$
  i.e. after k iterations, $\frac{1}{k}\sum_{i=1}^{k}(x_1^k, x_2^k, x_3^k, \lambda^k) \to (x_1^*, x_2^*, x_3^*, \lambda^*)$

Divergence

Divergence

Consider a situation where the orthogonality conditions are missing.

We examine the following linear homogeneous equation with three variables:

$$A_1x_2 + A_2x_2 + A_3x_3 = 0 \tag{12}$$

with the following assumptions:

- $A_i \in \mathcal{R}^3$ (i = 1, 2, 3) are column vectors
- matrix $[A_1, A_2, A_3]$ is non singular
- $x_i \in \mathcal{R}$

The unique solution to (12) is: $x_1 = x_2 = x_3 = 0$.

Regard (12) as a special case of the extended problem (4) with a null objective:

$$\begin{aligned} &\min 0(x_1) + 0(x_2) + 0(x_3) \\ &\text{s.t. } A_1x_1 + A_2x_2 + A_3x_3 = 0 \end{aligned} \tag{13}$$

Divergence

Applying the FOC to the Lagrange multipliers:

$$
\begin{cases}
-A_1^T\lambda^k + \beta A_1^T(A_1x_1^{k+1} + A_2x_2^k + A_3x_3^k) = 0, & \text{(14a)}\\
-A_2^T\lambda^k + \beta A_2^T(A_1x_1^{k+1} + A_2x_2^{k+1} + A_3x_3^k) = 0, & \text{(14b)}\\
-A_3^T\lambda^k + \beta A_3^T(A_1x_1^{k+1} + A_2x_2^{k+1} + A_3x_3^{k+1}) = 0, & \text{(14c)}\\
\beta(A_1x_1^{k+1} + A_2x_2^{k+1} + A_3x_3^{k+1}) + \lambda^{k+1} - \lambda^k = 0. & \text{(14d)}
\end{cases}
$$

Now setting $\mu^k := \lambda^k/\beta$,

$$
\begin{cases}
-A_1^T\mu^k + A_1^T(A_1x_1^{k+1} + A_2x_2^k + A_3x_3^k) = 0, & \text{(15a)}\\
-A_2^T\mu^k + A_2^T(A_1x_1^{k+1} + A_2x_2^{k+1} + A_3x_3^k) = 0, & \text{(15b)}\\
-A_3^T\mu^k + A_3^T(A_1x_1^{k+1} + A_2x_2^k + A_3x_3^{k+1}) = 0, & \text{(15c)}\\
(A_1x_1^{k+1} + A_2x_2^{k+1} + A_3x_3^{k+1}) + \mu^{k+1} - \mu^k = 0. & \text{(15d)}
\end{cases}
$$

And from (15a), we find:

$$
x_1^{k+1} = \frac{1}{A_1^TA_1}(-A_1^TA_2x_2^k - A_1^TA_3x_3^k + A_1^T\mu^k) \tag{16}
$$

Divergence

Substituting (16) into (15b-d), we arrive at the following system of equations:

$$\underbrace{\begin{pmatrix} A_2^T A_2 & 0 & 0_{1\times 3} \\ A_3^T A_2 & A_3^T A_3 & 0_{1\times 3} \\ A_2 & A_3 & I_{3\times 3} \end{pmatrix}}_{L} \begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} = \underbrace{\left[\begin{pmatrix} 0 & -A_2^T A_3 & A_2^T \\ 0 & 0 & A_3^T \\ 0_{3\times 1} & 0_{3\times 1} & I_{3\times 3} \end{pmatrix} - \frac{1}{A_1^T A_1} \begin{pmatrix} A_2^T A_1 \\ A_3^T A_1 \\ A_1 \end{pmatrix} (-A_1^T A_2 \,, -A_1^T A_3 \,, A_1^T) \right]}_{R} \begin{pmatrix} x_2^{k} \\ x_3^{k} \\ \mu^{k} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} = \underbrace{L^{-1} R}_{M} \begin{pmatrix} x_2^{k} \\ x_3^{k} \\ \mu^{k} \end{pmatrix}$$

So the iterative scheme is:

$$\begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} = M \begin{pmatrix} x_2^{k} \\ x_3^{k} \\ \mu^{k} \end{pmatrix} = \cdots = M^{k+1} \begin{pmatrix} x_2^{0} \\ x_3^{0} \\ \mu^{0} \end{pmatrix} \tag{17}$$

Note that matrix M is independent of β.

So convergence is achieved

$\Longleftrightarrow$ matrix mapping M is a contraction
$\Longleftrightarrow$ spectral radius $(\rho(M)) < 1$

Divergence

To construct a divergent example, look for an A s.t. $\rho(\mathrm{M}) > 1$, and s.t. any two coefficient matrices are **non-orthogonal**.

Consider:

$$A = (A_1, A_2, A_3) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix}$$

Computing the matrix M from before,:

$$M = L^{-1}R = \frac{1}{162}\begin{pmatrix} 144 & -9 & -9 & -9 & 18 \\ 8 & 157 & -5 & 13 & -8 \\ 64 & 122 & 122 & -58 & -64 \\ 56 & -35 & -35 & 91 & -56 \\ -88 & -26 & -26 & -62 & 88 \end{pmatrix}$$

Which admits the eigenvalue decomposition:

$$M = V\mathrm{Diag}(d)V^{-1}, \quad \text{where } d = \begin{pmatrix} 0.9836 + 0.2984i \\ 0.9836 - 0.2984i \\ 0.8744 + 0.2310i \\ 0.8744 - 0.2310i \\ 0 \end{pmatrix}$$

Note: $\rho(M) = |d_1| = |d_2| > 1$

Divergence

It remains to determine whether there are any real-valued non-convergent starting points.

Indeed, define the initial point (x_2^0, x_3^0, μ^0) as $V(:,1) + V(:,2) \in \mathbb{R}^5$, then:

$$\begin{aligned}
\begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} = \left[V\mathrm{Diag}(d)V^{-1}\right]^{k+1} \begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu^0 \end{pmatrix} &= V\mathrm{Diag}(d^{k+1})V^{-1} \begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu^0 \end{pmatrix} \\
&= V\mathrm{Diag}(d^{k+1}) \begin{pmatrix} 1 \\ 1 \\ 0_{3\times 1} \end{pmatrix} \\
&= V \begin{pmatrix} (0.9836 + 0.2984i)^{k+1} \\ (0.9836 - 0.2984i)^{k+1} \\ 0_{3\times 1} \end{pmatrix}
\end{aligned}$$

which is divergent.

Divergence - Theorem

Theorem 3.1. For the three-block convex minimization problem (4), there is an example where the direct extension of ADMM (5) is divergent for any penalty parameter $\beta > 0$ and for any starting point in a continuously dense half space of dimension 3.

In our example, set the initial point to be: $\begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu_1^0 \\ \mu_2^0 \\ \mu_3^0 \end{pmatrix} = V \begin{pmatrix} \alpha_1 \\ \alpha_1 \\ \alpha_2 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$

The points with $(\alpha_1 > 0, \alpha_2, \alpha_3) \in \mathcal{R}^3$ form a continuously dense half space of non-convergent starting points.

Extensions

Remark If all functions θ_i in (4) are assumed to be strongly convex and the penalty parameter β is restricted to a specific range determined the strong convex modulus of these functions, then the direct extension of ADMM is convergent.

[D. R. Han and X.M. Yuan, 2012]

Extensions

What about convergence for a strongly convex optimization model when the restriction on β is removed?

Consider the following strongly convex minimization problem:

$$\begin{aligned} \min\ & 0.05x_1^2 + 0.05x_2^2 + 0.05x_3^2 \\ \text{s.t. } & \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \end{aligned} \tag{18}$$

In a similar calculation to before, the spectral radius for the relevant matrix for (18) with $\beta = 1$ is $\rho(M) = 1.0087 > 1$. Thus again one can find a starting point such that the scheme diverges.

Theorem 4.1. For the model (4) with the strong convex assumption on its objective function, the direct extension of ADMM (5) is not necessarily convergent for all $\beta > 0$.

Conclusions

Conclusions

- The direct extension of the ADMM is not necessarily convergent for any given β.
- Given certain orthogonality conditions, convergence can be achieved.
- Even with strong convexity, the direct extension loses convergence for a given β.
- This justifies the algorithmic design and correction steps in recent papers to produce an algorithm with provable convergence for multi-block convex models.

Thank you!